

Economic Feasibility Calculator

Methodology and data record

Field	Detail
Project	Tasmanian Low Emissions Livestock Project
Prepared by	TasFarmers
Prepared for	Project record, supporting grant deed Milestone 6 reporting to the Department of State Growth
Date	23 July 2026
Version	1.0, initial record, describes the calculator as currently published
Status	Current

1. Purpose and grant deed requirement

This record documents how the Economic Feasibility Calculator works: its data sources, calculation logic and assumptions, so that the methodology behind every figure it produces can be checked and reproduced.

The calculator satisfies grant deed clause (ii), which requires *“development of a tool for livestock producers to calculate the economic feasibility of adoption of low emissions feed supplements, informed by the results of the Project.”*

The grant deed is a \$4,000,000 agreement with the Tasmanian Government, administered by the Department of State Growth, running from 19 June 2024 to 1 February 2029.

Two design choices flow directly from the clause: the tool is built for producers, not only as a funder-facing report, and every input to its calculations must trace back to actual trial data recorded by this project, not published industry averages.

2. Tool overview

The calculator is a live, self-contained web tool published on the Low Emissions Livestock Hub site, embedded on the Economic feasibility tool page, linked from the Resources page. A producer enters their own herd size, current weight, growth rate and cost of production, and the tool applies the SeaFeed-versus-control difference measured in the trial to date to estimate what adopting SeaFeed could mean for their operation.

It offers two view modes, selected by the producer:

- reach a target weight: compares how many days business-as-usual and SeaFeed-supplemented cattle take to reach the same nominated sale weight; and
- feed for a set period: compares the final weight each pathway reaches over the same nominated number of days.

Both modes draw on the same underlying trial coefficients, described in section 5; they present different lenses on the same figures.

3. Data sources and lineage

Everything the calculator shows is derived, directly or indirectly, from two live data sources: the project's own Economics data table and a public national livestock price indicator. This section traces each figure back through the project's data pipeline to where it is first recorded.

3.1 Lick block cost and head count

Feed cost and head count are recorded through Survey123 at each trial site and logged in the Lick Block Data layer. The Economics Pipeline sums recorded cost to farm and calculates a representative head count, the average of recorded head counts, for each farm and treatment (SeaFeed or control) combination.

3.2 Weight gain

Weight gain per head is calculated from the Productivity item: the earliest and most recent recorded average weight for each farm and treatment, giving total gain over the whole period tracked so far.

3.3 Days of data

Days of data counts the number of days with a recorded methane abatement reading in the Emissions item, for each farm and treatment. It measures how long trial monitoring has been running for that group, not how often weight has been recorded.

3.4 Cost per head and cost per kilogram of gain

The Economics Pipeline calculates the following for each farm and treatment, and for a trial-wide aggregate:

```
cost per head = cumulative cost ÷ representative head count  
cost per kilogram of gain = cost per head ÷ weight gain per head
```

These figures, alongside days of data, are written daily to the Economics data table, appended rather than overwritten, so cost-effectiveness can be tracked improving over time as fixed costs amortise against growing weight gain.

3.5 The trial-wide aggregate row

Alongside a row for each participating farm, the Economics Pipeline builds one head-count-weighted aggregate row per treatment (farm_name = "All Farms (Trial Aggregate)"), pooling every farm's cost and head count into a single trial-wide figure. The calculator queries only these two aggregate rows, never a named farm's individual row, see section 4.

3.6 Market price reference

The calculator's suggested sale price is auto-filled from the Meat & Livestock Australia (MLA) Statistics API, using the National Heavy Steer Indicator (cents per kilogram liveweight), the closest published national reference to a finished steer at a typical target sale weight. It is a national figure, not Tasmania-specific, Tasmanian saleyards do not report sufficient volume for their own daily indicator, and is always shown as a starting point the producer can override with local knowledge.

4. Data governance and privacy

The calculator queries only the trial-wide aggregate rows of the Economics data table, filtered explicitly to `farm_name = "All Farms (Trial Aggregate)"` in the query itself, never the per-farm rows also held in the same table. This is a deliberate data-sensitivity safeguard: farm-level and producer-identifying performance data is not exposed in this producer-facing tool, consistent with the project's data protocols, which follow the National Farmers' Federation Farm Data Code.

5. Calculation methodology

5.1 Step one: deriving the trial differential

The calculator does not use either treatment group's absolute figures directly. Instead, it calculates the difference between the SeaFeed and control aggregate rows, isolating the effect of SeaFeed from general trial conditions that affect both groups equally.

```
incremental cost per head per day = (SeaFeed cost per head - control cost per head)
÷ days of data
growth rate uplift (kg/day) = (SeaFeed weight gain per head - control weight gain
per head) ÷ days of data
```

Both figures are recalculated every time the calculator loads, from whichever aggregate rows are most current in the Economics data table.

5.2 Step two: applying the differential to the producer's own mob

The growth rate uplift and incremental cost are applied on top of the producer's own entered growth rate and cost of production, not the trial's absolute figures. This personalises the result to the producer's own operation rather than assuming their cattle behave exactly like the trial's participating herds.

The producer supplies:

- growth rate, entered directly in kilograms per day, or derived from current weight, target weight and the typical number of days to reach it; and
- cost of production, entered as dollars per head per day, or dollars per kilogram of liveweight, converted internally using the producer's own growth rate.

```
SeaFeed growth rate = producer's growth rate + growth rate uplift
SeaFeed cost per head per day = producer's cost of production + incremental cost per
head per day
```

5.3 View: reach a target weight

Let gap = target weight – current weight (minimum one kilogram).

```
days (business as usual) = gap ÷ business-as-usual growth rate
days (SeaFeed) = gap ÷ SeaFeed growth rate
total cost (each pathway) = cost per day × days for that pathway
cost per kilogram (each pathway) = total cost ÷ gap
net saving across the herd = (total cost business as usual - total cost SeaFeed) ×
herd size
revenue per head = target weight × sale price (identical for both pathways)
gross margin (each pathway) = (revenue per head - total cost) × herd size
```

Because both pathways sell at the same target weight, revenue per head is identical either way, so the difference between the two gross margin figures always equals the net saving above. Showing both absolute figures alongside the saving gives scale and context, not new arithmetic.

5.4 View: feed for a set period

Let d = the nominated feeding period, in days (minimum one).

```
weight gained (each pathway) = growth rate for that pathway × d
total cost (each pathway) = cost per day for that pathway × d
cost per kilogram (each pathway) = total cost ÷ weight gained
extra weight = weight gained (SeaFeed) – weight gained (business as usual)
extra revenue = extra weight × sale price
extra cost = total cost (SeaFeed) – total cost (business as usual)
net saving across the herd = (extra revenue – extra cost) × herd size
gross margin (each pathway) = ((current weight + weight gained) × sale price – total cost) × herd size
```

Unlike the target-weight view, the two pathways finish at different weights on the same day, so sale price genuinely changes each pathway's revenue, rather than only affecting the difference between them.

5.5 Gross margin, and why it does not reach zero when cost of production equals sale price

Gross margin, as reported by the calculator, only charges cost against the weight gained during the period being modelled. No cost is charged against the animal's starting weight, because that weight is treated as already paid for before the modelled period begins. As a result, gross margin includes the value of the animal's full starting weight at the sale price, with no cost counted against it, so it remains positive even when cost of production exactly equals sale price. This is intentional: it isolates the economics of the finishing period being modelled and is not a measure of whole-of-life farm profit.

Worked example: a 235kg store animal, with cost of production entered equal to sale price, finished to 500kg, still returns a positive gross margin, because roughly 235kg of the 500kg sale weight is charged no cost at all in this model.

6. Data confidence, limitations and assumptions

The calculator carries the following limitations:

- the trial differential is drawn from every participating farm pooled together, and does not currently vary by breed, feedlot versus pasture system, or individual farm conditions;
- days of data reflects how long emissions monitoring has been running, and remains low in the trial's early stages, so early results should be read as preliminary and will firm up as monitoring continues;
- the suggested sale price is a national indicator, not a Tasmanian or saleyard-specific figure, and does not account for seasonal price variation, transport, or other on-farm costs;
- all mob inputs, herd size, current weight, growth rate and cost of production, are entered by the producer and are not independently verified by the calculator; and
- the tool does not constitute financial advice; its own footnote directs producers to their accountant or farm business adviser for a full assessment of their operation.

The calculator's status indicator shows the number of days of trial data behind the current result at all times, and is designed to always display real trial figures, including periods where SeaFeed underperforms the control group, rather than withholding results below a chosen threshold.

7. Update cadence

The Economics Pipeline runs daily, replacing that day's snapshot in the Economics data table in a re-run-safe append pattern: any existing snapshot from the same day is cleared before the new one is added. The calculator queries the most current aggregate rows each time it loads, so results improve automatically as trial monitoring continues and as further farms come online, with no manual update required to the calculator itself.

8. Document control and version history

Version	Date	Change
1.0	23 July 2026	Initial audit record, describing the calculator as published. The methane abatement co-benefit was removed from the calculator prior to this version, to be developed as a separate illustrative carbon value tool.

Appendix A: field and formula reference

Calculator field	Formula	Source
Incremental cost per head per day	$(\text{SeaFeed cost per head} - \text{control cost per head}) \div \text{days of data}$	Economics data table, aggregate rows
Growth rate uplift (kg/day)	$(\text{SeaFeed weight gain per head} - \text{control weight gain per head}) \div \text{days of data}$	Economics data table, aggregate rows
Cost per kilogram (target weight view)	$\text{total cost} \div (\text{target weight} - \text{current weight})$	Calculated at run time from producer inputs
Cost per kilogram (duration view)	$\text{total cost} \div \text{weight gained over the period}$	Calculated at run time from producer inputs
Net saving across the herd	see sections 5.3 and 5.4 for the view-specific formula	Calculated at run time from producer inputs
Gross margin (each pathway)	see sections 5.3 and 5.4 for the view-specific formula	Calculated at run time from producer inputs
Suggested sale price	MLA National Heavy Steer Indicator, most recent reading	MLA Statistics API

Appendix B: data source endpoints

Source	Endpoint / identifier
Economics data table	LEL_Economics_Data FeatureServer, item c84581316ebf4802b38693fd0aac605a
Lick block data	item ccdeb35b6d254baf8e9977a175aa3605
Emissions data	item 9fbc1b9596b24ad5aac0623221d60391
Productivity data	item 3dcaf48fff994610922fbb78afb90b4d
Market price reference	MLA Statistics API, api-mlastatistics.mla.com.au, indicator ID 4 (National Heavy Steer Indicator)

Item identifiers refer to hosted feature layers and tables in the project's ArcGIS Online organisation.